Multiplicative inverses in GF(2^n) and the AES S-box

Recall that in GF(2^8) the field elements are multiplied and added modulo $m(x) = x^8 + x^4 + x^3 + x + 1$, which can be represented as *m(x)* = {11B} = 100011011.

The substitution values ($b_7b_6b_5b_4b_3b_2b_1b_0$) found in the AES S-box are derived by computing the multiplicative inverse ($a_7a_6a_5a_4a_3a_2a_1a_0$) of the initial 8-bit value, and then computing each bit of the substitution value according to:

$$b_i = \left(a_i + a_{i+4} + a_{i+5} + a_{i+6} + a_{i+7} + c_i\right) \bmod 2$$

where $c_7c_6c_5c_4c_3c_2c_1c_0$ = 01100011

note: coefficients are reduced mod2 and subscripts are reduced mod8.

Example: Compute the AES S-box value for {53}.

{53} is 0101 0011

The inverse of 0101 0011 can be found using the Extended Euclidean Algorithm (standard, fast, or classic). The computations shown here are in binary/hex (but they can also be done using the 256 polynomial elements of GF(2^8)…). Using **EEA_fast**:

binary	hex	*s*	*t*
1 0001 1011	{11B}	1	0
0101 0011	{53}	0	1
1 0001 1011	{11B}		
1 0100 1100	-{04}{53}		
101 0111			
101 0011	-{01}{53}		
100	{11B}-**{05}**{53}={04}	1-**{05}**0=1	0-**{05}**1={05}
0101 0011	{53}		
100 0000	-{10}{04}		
1 0011			
1 0000	-{04}{04}		
11	{53}-**{14}**{04}={03}	0-**{14}**1={14}	1-**{14}**{05}={45}
100	{04}		
110	-{02}{03}		
10			
11	-{01}{03}		
1	{04}-**{03}**{03}={01}	1-**{03}**{14}={3D}	{05}-**{03}**{45}={CA}

Note: You can also compute the inverse of {53} in $GF(2^8)$ using **EEA_classic**:

100011011		*a*={11B}
1010011		*b*={53}
100011011		
101001100	{04}x1010011	*a* – **{05}***b* = 100
1010011	{01}x1010011	*a* + {05}*b* = 100
100	**{05}**x1010011	
1010011		
1000000	{10}x100	(*b*) – **{14}**(*a*+{05}*b*) = 11
10000	{04}x100	*b* + {14}*a* + {44}*b* = 11
11	**{14}**x100	{14}*a* + {45}*b* = 11
100		
110	{02}x11	(*a* + {05}*b*) – **{03}**({14}*a* + {45}*b*) = 1
11	{01}x11	*a* + {05}*b* +{3C}*a* + {CF}*b* = 1
1	**{03}**x11	{3D}*a* + {CA}*b* = 1

Therefore, {3D}{11B}-{CA}{53}={01}. {CA}=1100 1010, and {CA}{53} is found by:

```
10100110000000      {53}1000 0000
 1010011000000      {53}0100 0000
     1010011000     {53}0000 1000
       10100110     {53}0000 0010
11111101111110
```

Reducing {CA}{53}=11111101111110 by *m(x)*=100011011 yields 1:

```
11111101111110
100011011
 1110000011110
 100011011
  110110101110
  100011011
   10101110110
   100011011
      100011010
      100011011
              1
```

Finding the S-box value for {53}:

{CA}= $a_7a_6a_5a_4a_3a_2a_1a_0$ $c_7c_6c_5c_4c_3c_2c_1c_0$ (the 'c' values never change)

1 1 0 0 1 0 1 0 0 1 1 0 0 0 1 1

$$b_i = \left(a_i + a_{i+4} + a_{i+5} + a_{i+6} + a_{i+7} + c_i\right) \bmod 2$$

$b_0 = \left(a_0 + a_{0+4} + a_{0+5} + a_{0+6} + a_{0+7} + c_0\right) \bmod 2 =$ 0+0+0+1+1+1 mod2 = 1

$b_1 = \left(a_1 + a_{1+4} + a_{1+5} + a_{1+6} + a_{1+7} + c_1\right) \bmod 2 =$ 1+0+1+1+0+1 mod2 = 0

$b_2 = \left(a_2 + a_{2+4} + a_{2+5} + a_{2+6} + a_{2+7} + c_2\right) \bmod 2 =$ 0+1+1+0+1+0 mod2 = 1

$b_3 = \left(a_3 + a_{3+4} + a_{3+5} + a_{3+6} + a_{3+7} + c_3\right) \bmod 2 =$ 1+1+0+1+0+0 mod2 = 1

$b_4 = \left(a_4 + a_{4+4} + a_{4+5} + a_{4+6} + a_{4+7} + c_4\right) \bmod 2 =$ 0+0+1+0+1+0 mod2 = 0

$b_5 = \left(a_5 + a_{5+4} + a_{5+5} + a_{5+6} + a_{5+7} + c_5\right) \bmod 2 =$ 0+1+0+1+0+1mod2 = 1

$b_6 = \left(a_6 + a_{6+4} + a_{6+5} + a_{6+6} + a_{6+7} + c_6\right) \bmod 2 =$ 1+0+1+0+0+1 mod2 = 1

$b_7 = \left(a_7 + a_{7+4} + a_{7+5} + a_{7+6} + a_{7+7} + c_7\right) \bmod 2 =$ 1+1+0+0+1+0 mod2 = 1

The S-box substitution value ($b_7b_6b_5b_4b_3b_2b_1b_0$) for {53} is 1110 1101 or {ED}.

Exercises:

1. In GF(2^3) with $m(x) = x^3 + x + 1$:
   a) Find the multiplicative inverse of (110) using the Extended Euclidean Algorithm.
   b) Find the multiplicative inverse of (010) using the Extended Euclidean Algorithm.
   c) Verify that the multiplicative inverse of (111) is (100) by reducing (111)x(100) by *m(x).*

2. In GF(2^5) with $m(x) = x^5 + x^2 + 1$:
   a) Find the multiplicative inverse of (11001) using the Extended Euclidean Algorithm.
   b) Verify your result from part a) by reducing the product of (11001) and its inverse by *m(x)*.

3. If the inverse of a GF(2^8) element is {F0}, what is its AES S-box value? What is the element?

4. Compute the AES S-box value for {42}. Show the computation of the <u>inverse</u> and <u>each bit</u> of the final value.

Answers:

1.

a) 011
b) 101
c) 11100 reduced by 1011 is 1

2.

a) 1010
b) 11111010 reduced by 100101 is 1

3. value={39}, element={5B}

4. inverse is 110111, value is 00101100